Vomals – Voronoi Analysis of Categorical Data

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TBD

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Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All Rmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at https://github.com/deleeuw/voronoi

1 Introduction

2 Homogeneity Analysis

Homogeneity analysis, a.k.a. *Multiple Correspondence Analysis*, can be introduced in many different ways. We refer to Guttman (1941), Tenenhaus and Young (1985), Bekker and De Leeuw (1988), Gifi (1990), Greenacre (1991), Le Roux and Rouanet (2010), De Leeuw (2023) for a bouquet of discussions and interpretations of the equations and algorithms defining the technique.

For our purposes a geometric interpretation using *star plots* seems the most natural one. We have n objects measured by m categorical variables. Variable j has k_j categories, which are the different values the variable can assumes for the n objects. Suppose the objects are mapped into n points of Euclidean space $\mathbb{R}^p$. For each variable we can construct an additional k_j points in the same space by computing for each category the centroids of all object-points in that category. Thus each variable defines its own plot. We then draw lines from the n object points to "their" category point and we obtain k_j "stars". Homogeneity analysis maps the objects into space in such a way that the stars are as small as possible, where the size of the star is measured as the sum of squares of the distances between each category points and "its" object points (summed over categories and over variables). In order to avoid the trivial solution in which all objects are mapped into the same point homogeneity requires that the *configuration* of object points (an $n \times p$ matrix) is column-centered and orthonormal.

There are a huge number of variations possible on the same theme. Instead of the centroids we could use the Weber points to define our stars. Instead of Euclidean distance we could use any other distance on $\mathbb{R}^p$. Instead of the stars we could use the convex hulls or the minimum spanning trees, or any other way to visualize and quantify the compactness of the point clouds of each category. Any one of these choices defines a technique that could be called homogeneity analysis, but in this paper we mean the classical choice that uses stars, squared Euclidean distance, and orthonormal object points.

The key mathematical object in homogeneity analysis is the *indicator matrix*, which is a classical way to code categorical data. For a variable j with k_j categories it is an $n \times k_j$ binary matrix G_j in which each row i contains a single element equal to one, indicating into which category the object i maps for variable j. The cross product $D_j := G_j'G_j$ is diagonal, with the *marginal frequencies* of the categories on the diagonal. For two different variables the crossproduct $C_{j\ell} = G_j'G_\ell$ is the $k_j \times k_\ell$ *cross table* or *contingency table* of the two variables.

The G_j can be used as weights in the traditional "stress" multidimensional scaling loss function, defined as

$$\sigma(X;Y_1,\cdots,Y_m):=\sum_{j=1}^m\sum_{i=1}^n\sum_{\ell=1}^{k_j}g_{i\ell}^j(\delta_{i\ell}^j-d(x_i,y_\ell^j))^2.$$

Here the dissimilarity $\delta_{i\ell}^j=0$ if $g_{i\ell}^j=1$. If $g_{i\ell}^j=0$ the dissimilarity is arbitrary, because that particular triple (i,j,ℓ) does not contribute to the loss function. Since $\delta_{i\ell}^j g_{i\ell}^j=0$ for all (i,j,ℓ) we have

$$\sigma(X;Y_1,\cdots,Y_m)=\sum_{j=1}^m\sum_{i=1}^n\sum_{\ell=1}^{k_j}g_{i\ell}^j\{d(x_i,y_\ell^j)\}^2.$$

Define

$$\sigma_\star(X):=\min_{Y_1,\cdots,Y_m}\sigma(X;Y_1,\cdots,Y_m).$$

The minimum over the Y_j is attained at the centroids $\widehat{Y}_j:=D_j^{-1}G_j'X$, and if we substitute these optimal $\widehat{Y}_j$ in … we see that $\sigma_\star(X)$ is indeed equal to our measure of the total size of all stars. Minimizing $\sigma_\star(X)$ over $X'X=I$ is homogeneity analysis, and computing the solution turns out to be a straightforward eigenvalue-eigenvector problem.

3 Voronoi Analysis

$$\sigma(X,Y,\Delta)=\sum_{i=1}^n\sum_{j=1}^m\|\delta_{ij}-d_{ij}(X,Y)\|^2$$

If $g_{i\ell}^j=1$ then $\{\delta_{ij}\}_\ell\leq\{\delta_{ij}\}_\nu$ for all $\nu=1,\cdots,k_j$.

$$\delta_{ij}'J\delta_{ij}=1$$

Majorization

$$A_{ij}:=\begin{bmatrix}e_ie_i'&-e_ie_j'\\-e_je_i'&e_je_j'\end{bmatrix}$$

$$Z:=\begin{bmatrix}X\\Y\end{bmatrix}$$

$$d_{ij}(X,Y)=\text{tr }Z'A_{ij}Z$$

$$V:=\sum_{i=1}^n\sum_{j=1}^m A_{ij}=\begin{bmatrix}mI&-ee'\\-ee'&nI\end{bmatrix}$$

If we define

$$V_1:=(n+m)^{-1}\begin{bmatrix}\frac{m}{n}E&-E\\-E&\frac{n}{m}E\end{bmatrix}, \tag{1}$$

$$V_2:=\begin{bmatrix}J_n&0\\0&0\end{bmatrix}, \tag{2}$$

$$V_3:=\begin{bmatrix}0&0\\0&J_m\end{bmatrix}, \tag{3}$$

then

$$V=(n+m)V_1+mV_2+nV_3$$

Matrices V_1, V_2, and V_3 are symmetric projectors on orthogonal subspaces. It follows that V^+, the Moore-Penrose inverse3 of V, is

$$V^+=\frac{1}{n+m}V_1+\frac{1}{m}V_2+\frac{1}{n}V_3$$

$$B(X,Y):=\begin{bmatrix}B_{11}(X,Y)&B_{12}(X,Y)\\B_{21}(X,Y)&B_{22}(X,Y)\end{bmatrix}$$

Now $V_1B(X,Y) = 0$ and thus

$$V^+B(X,Y) = \frac{1}{m}V_2B(X,Y) + \frac{1}{n}V_3B(X,Y)$$

4 The Cone

Consider the cone of all vectors x in $\mathbb{R}^n$ for which $x_i \leq x_1$ for all $i \neq 1$ and $e'x = 0$. The extreme rays of the cone will be the $n-1$ vectors of the form $\alpha e_i + \beta(e - e_i)$ with $i \neq 1$. Thus element i is equal to α and all other elements (including x_1) are equal to beta. Thus we must have $\alpha > \beta$. The sum of the elements is $\alpha + (n-1)\beta$, which must be zero. Thus $\alpha = -(n-1)\beta$, which implies $\beta < 0$ and $\alpha > 0$. The extreme rays are positive multiples of the vectors $e_i - n^{-1}e$ with $i \neq 1$ and the cone is the set of all weighted sums with non-negative weights of these $n-1$ vectors. Here is a small example.

```
a <- -cbind(1, -diag(4))
b <- (diag(5)-(1 / 5))[, -1]
print(a)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]   -1    1    0    0    0
[2,]   -1    0    1    0    0
[3,]   -1    0    0    1    0
[4,]   -1    0    0    0    1
```

```
print(b)
```

```
     [,1] [,2] [,3] [,4]
[1,] -0.2 -0.2 -0.2 -0.2
[2,]  0.8 -0.2 -0.2 -0.2
[3,] -0.2  0.8 -0.2 -0.2
[4,] -0.2 -0.2  0.8 -0.2
[5,] -0.2 -0.2 -0.2  0.8
```

```
print(a %*% b)
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    0    0    0
[2,]    0    1    0    0
[3,]    0    0    1    0
[4,]    0    0    0    1
```

For later reference the sum of squares of the vectors $e_i - n^{-1}e$ is $(n-1)/n$, so the vectors $\sqrt{n/(n-1)}(e_i - n^{-1}e)$ are the vectors of length 1 on the extreme rays. The average of the extreme rays is

5 Normalized Monotone Regression

Suppose y is a vector with n elements. The problem we study is to minimixe the sum of squares $\sigma(x) := \|x-y\|^2$ over $x \in K$ and $x'Jx = 1$. Here

- K is the cone of vectors with $x_1 \leq x_2 \leq \cdots \leq x_n$,
- J is the centering matrix $J := I - n^{-1}ee'$, where
- e is the vector with all n elements equal to one.

Change variables to $\tilde{x} = Jx$. Then $x'Jx = \tilde{x}'\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|^2$ over β and $\tilde{x} \in K$, with $e'\tilde{x} = 0$ and $\tilde{x}'\tilde{x} = 1$. The minimizer for β is $\overline{y} := n^{-1}e'y$, and thus the problem becomes minimizing $\|\tilde{x} - Jy\|^2$ over $\tilde{x} \in K$, $\tilde{x}'\tilde{x} = 1$, and $e'\tilde{x} = 0$.

Forget about the constraint $e'\tilde{x} = 0$. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $M(Jy)/\|M(Jy)\|$, which automatically satisfies $e'\tilde{x} = 0$. Thus the solution of our original problem is

$$\hat{x} = \frac{M(y-\overline{y})}{\|M(y-\overline{y})\|} + \overline{y}.$$

$M(y-\overline{y}) = M(y) - \overline{y}$ and thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 + n\overline{y}^2 - 2\overline{y}e'M(y)$ nor $e'M(y) = e'y = n\overline{y}$. Thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 - n\overline{M(y)}^2$

$$\hat{x} = \frac{M(y) - \overline{y}e}{\|M(y-\overline{y})\|} + \overline{y}e = \frac{1}{\|M(y-\overline{y})\|}M(y) + \left\{1 - \frac{1}{\|M(y-\overline{y})\|}\right\}\overline{y}e$$

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